

SMART INDIA HACKATHON · COMPLETE PLAYBOOK

How to Win *SIH.*

The complete system — from registration to the grand finale.

49,892

teams registered in SIH 2024

1,405

teams reached the grand finale

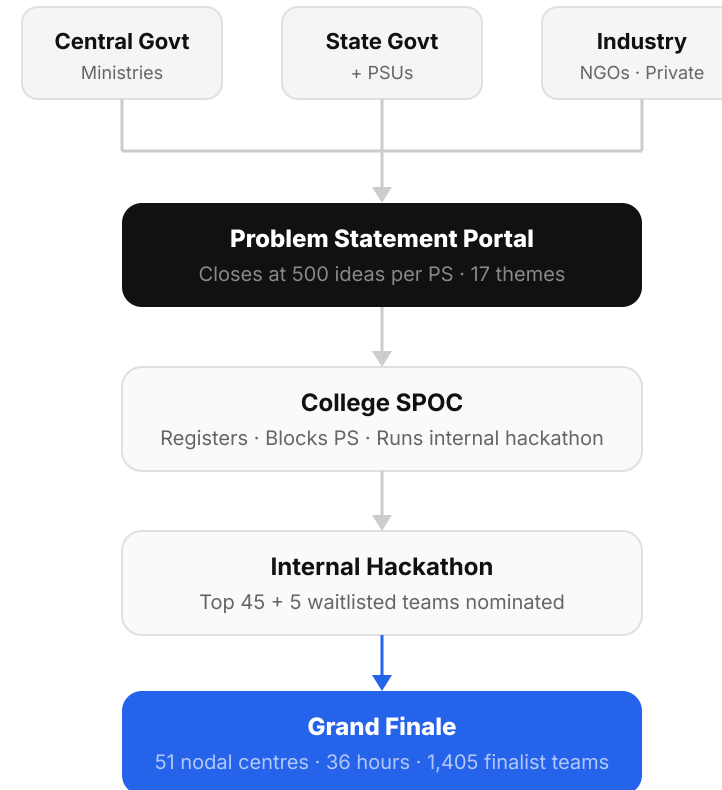
₹1L

prize per winning team

SIH is a government pipeline, not just a competition.

Ministries and PSUs submit real, unsolved problems. Student teams build solutions. Winners get funding and implementation support. The best solutions can end up in actual government systems.

SIH 2024 SCALE 57,378 ideas submitted · 51 nodal centres · 17 themes · Central + State ministries + PSUs



You lose at Stage 2 and 3 more often than at the finale.

1

PS Portal Opens

Ministries publish problems. Every PS has a live counter. When it hits **500 idea submissions** — it freezes. Popular PSs (ISRO, MeitY) can freeze in days.

2

SPOC Registers Your College

Your college's Single Point of Contact registers on the portal and blocks a PS. If your SPOC misses the deadline — your team cannot submit. This is Stage 2 death for many colleges.

3

Internal Hackathon

Your college runs its own hackathon. Your SPOC nominates a maximum of **45 teams + 5 waitlisted** to the portal. You must be in that list to submit your idea.

4

Idea Submission

Team leader submits your idea on the SIH portal. One team can apply to a **maximum of 2 PSs**. The submission format matters — judges score it before seeing your prototype.

5

Expert Evaluation

National-level jury evaluates all submissions on **10 official criteria**. A small number of teams per PS are shortlisted for the finale. The exact count varies by PS, but it's always a very small fraction.

6

Grand Finale

Offline at 51 nodal centres across India. 36 hours. **₹1 lakh prize** per winning team per PS. 1,405 finalist teams competed in SIH 2024.

Read the PS like a detective, not like a developer.

Ministries don't want an app. They want a **specific problem solved** within real constraints — connectivity, literacy, policy, budget. Most teams miss this completely.

REAL PS FROM SIH 2024

SIH1707 · GAIL · Ministry of Petroleum & Natural Gas

"Geolocation-Based Attendance Tracking Mobile Application"

Won by Team Coding_Wizard · Bhilai Institute of Technology, Durg

THE TEST Can you name the exact user, their specific pain, and what "solved" looks like for them?

HOW TEAM CODING_WIZARD WOULD'VE READ SIH1707

- WHO** **GAIL field employees**
Workers deployed across multiple off-site locations, not at a fixed desk
- PAIN** **Manual attendance is unreliable**
Paper registers, proxy signing, no real-time visibility for managers
- ROOT** **No location-aware system in place**
Existing HR tools don't handle geofencing for mobile workers
- WIN** **What the PS actually asked for**
Geofencing (200m radius) + real-time sync + manual check-in for off-site work + leave management integration

Lesson: The winning team didn't just build a location tracker. They mapped every constraint — off-site scenarios, connectivity gaps, manual override cases — into specific features. That's what reading a PS deeply means.

The best PS for your team is rarely the most famous one.

✗ HOW TEAMS PICK THE WRONG PS



"It's from ISRO or DRDO"

High-profile orgs attract hundreds of submissions. The 500-slot cap fills up fast. More competition, same number of finalist spots.



"We want to use AI / Blockchain"

Tech-first thinking. You pick a PS to match your tech stack — not to solve the ministry's actual problem. Judges can tell.



"Satellite imagery / hardware demo"

Hardware-heavy PSs need physical prototypes. Almost impossible to build, test, and demo reliably in 36 hours from scratch.

✓ HOW WINNERS PICK THEIR PS



Your team knows this domain

Someone on your team has real exposure — family member in that field, project, internship, or genuine interest. Context beats cleverness.



You can demo it in 5 minutes

A complete user flow — input → process → result — on a phone or laptop, with no special hardware. If you can't demo it cleanly, don't pick it.



Real data exists for it

Open government datasets, data.gov.in, NIC portals, or synthetic data you can pre-load. No data = no credible demo.

Each team can submit ideas for **max 2 PSs**. Each PS closes after **500 submissions**. Your college can nominate **max 50 teams** (45 + 5 waitlisted).

Same finalist slots. Fewer competitors.

Every PS gets a fixed number of finalist teams — regardless of how many teams submitted. A PS from a smaller organisation with fewer submissions means **the same opportunity with less competition.**

The math is obvious once you see it. Most teams never look at it this way.

THE RULE Optimise for: $\text{fit} \times \text{execution quality} \div \text{competition}$. Not for prestige or trending tech.

HOW TO COMPARE TWO PSS BEFORE PICKING

QUESTION TO ASK	PS A (FAMOUS ORG)	PS B (SMALLER ORG)
Does your team know this domain?	No	Yes
Can you demo it in 5 minutes?	Unclear	Yes
Is real data available?	Maybe	Yes — data.gov.in
Slots remaining when you check?	Very few	Many
Right choice?	✗	✓

REAL EXAMPLE FROM SIH 2024

SUPERNOVAZ

Ministry of AYUSH · PS SIH1556

Won by solving a digitisation problem for traditional medicine. Not glamorous. Not trending. But they understood the domain, built a clean solution, and won.

6 members. 6 different jobs. Not 6 coders.

The official rules: exactly 6 members, same college, at least 1 female member. You can have up to 2 mentors at the finale — but no two students from different colleges.

6

team members, all from the same college

≥1

female member — mandatory

2

optional mentors at the finale

2

PSs maximum per team

THE MOST MISSED ROLE

The presenter. Every team has 5 coders. Almost no one has a dedicated person who owns the pitch and rehearses it 20 times.

01

Product Lead

Owns the problem statement. Decides what gets built. Keeps scope from expanding. Has final say on feature cuts.

02

Backend Engineer

APIs, database, authentication, business logic. Builds the engine the demo runs on.

03

Frontend / UI

User flows, screens, interactions. What the judge actually sees and touches.

04

Data / AI

Model selection, integration, evaluation metrics, fallback logic. If AI is in the solution — this person owns it completely.

05

Researcher

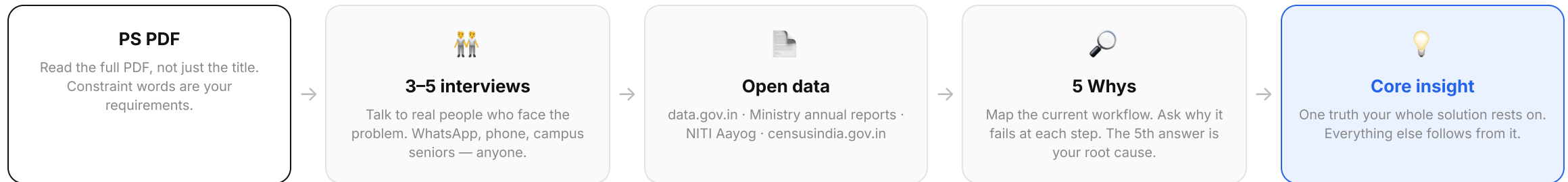
User interviews, government reports, domain knowledge. Grounds every claim in evidence. Judges ask "how do you know?" — this person has the answer.

06

Presenter / Storyteller

Demo script, pitch deck, Q&A preparation. Practices 20+ times. The most confident speaker on the team — not the smartest engineer.

Every winning team can trace every decision back to a real user truth.



THE SENTENCE EVERY TEAM MUST BE ABLE TO SAY OUT LOUD, WITH REAL SPECIFICS:

"[User type] currently faces [specific pain] because [root cause]. We know this because [evidence — report name, interview count, data source]. Our solution addresses it by [specific mechanism]."

If you can't fill this with real names and real sources — you are not ready to build. Go interview someone first.

One user journey. End to end. Working perfectly.

Not 10 features. Not a full product. The smallest version that proves your core value — completely, reliably — and can be demoed in 5 minutes without anything breaking.

BUILD

The core flow. Works end-to-end.
No shortcuts.

SIMULATE

Pre-loaded data. Cached AI outputs.
Realistic mock responses.

DEFER

Admin panel. Analytics. Multi-language. Reports. After demo.

CUT

Anything you can't reliably demo. No exceptions.

USING SIH1707 (GAIL ATTENDANCE) AS THE EXAMPLE

✗ WHAT WEAK TEAMS BUILD IN 36 HOURS

Employee login + profile page

GPS check-in (sometimes works)

Admin dashboard with attendance charts

Leave management module

Push notification system

Payroll integration (half-built)

3 of 6 features work on demo day. GPS crashes twice.

✓ WHAT WINNING TEAMS BUILD

Open app → auto-detect location → inside geofence → one-tap check-in → manager gets real-time notification → employee sees confirmation

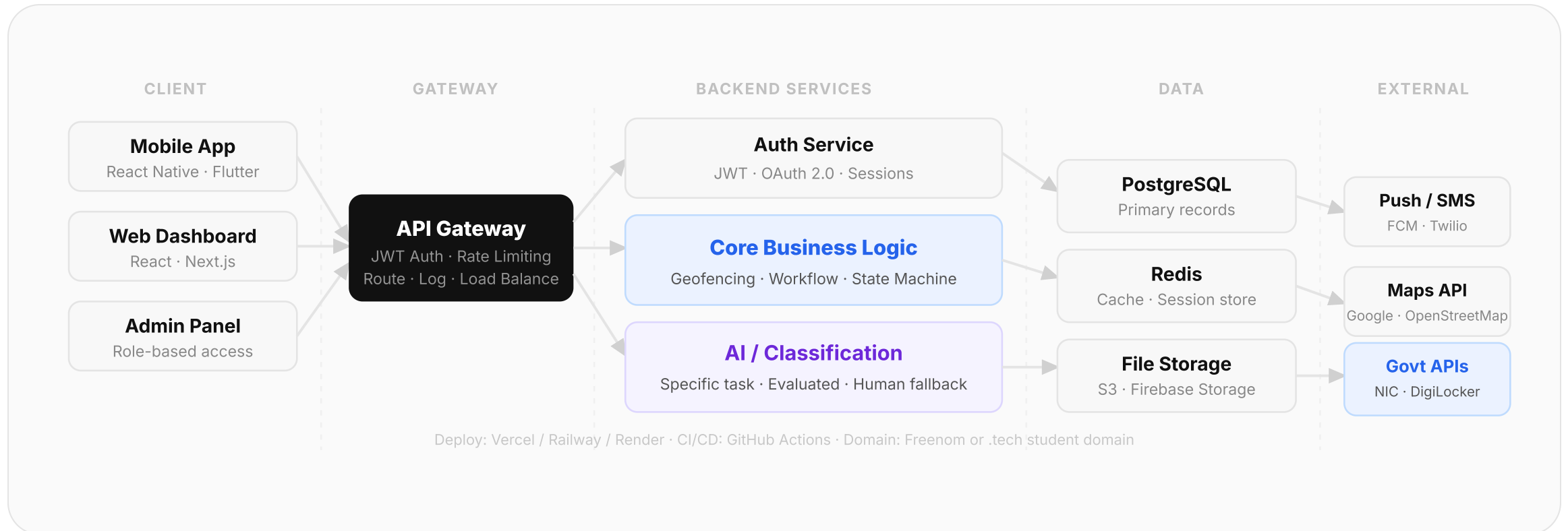
Off-site scenario: manual check-in with reason → pending approval → manager approves in 1 tap

Two flows. Both complete. Both reliable. Demo in 4 minutes.

THE LOCK RULE

No new features after Hour 16 of the finale. The next 20 hours are for polish, practice, and rest.

A clean architecture diagram tells the judge: "We know how to actually ship this."



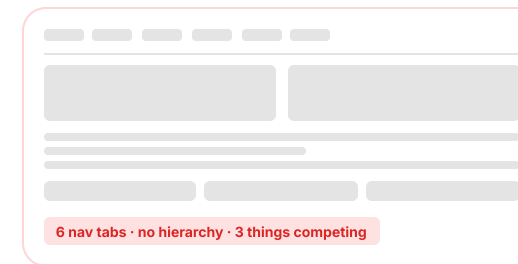
Judges will ask: what DB, how do you handle scale, what if a service fails? Know the answer for every box. The AI box especially — name the model, the task, the accuracy metric, and the fallback.

Your demo decides in the first 60 seconds.

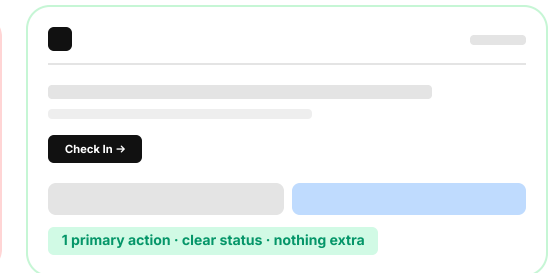
Judges form a first impression the moment your screen comes on. A cluttered dashboard with 8 tabs signals "we built a lot and don't know what matters." A clean screen with one action signals "we understand our user."

- Design the 5-step demo flow first, then build toward it
- Pre-load realistic, named demo data — never show an empty state to a judge
- Test on a device that isn't your own laptop
- One person runs the demo. Everyone else is quiet unless asked.
- Keep a screen recording of a perfect run — ready as backup

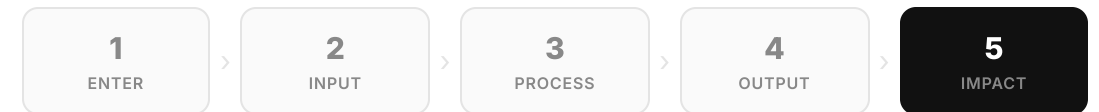
✗ WHAT LOSES



✓ WHAT WINS



THE 5-STEP DEMO STRUCTURE



One job. Measured. With a fallback.

Judges at SIH have seen hundreds of teams claim "AI-powered everything." What actually impresses them: a team that defines exactly what the AI does, shows how they measured it, and has thought about what happens when it's wrong.

THE 5 QUESTIONS JUDGES WILL ASK ABOUT YOUR AI

- 1 What model are you using and why this one specifically?
- 2 What data was it trained on or prompted with?
- 3 What's your accuracy — on what test set?
- 4 What happens when the AI output is wrong or uncertain?
- 5 Why is AI better than a simple rule-based system for this task?



Classification

Auto-route a complaint to the correct department based on text description



Summarisation

Condense a 3-page report into a 3-line summary for a busy officer



Anomaly Detection

Flag unusual patterns in attendance data that a rules engine would miss



Document Extraction

Pull structured fields from scanned government forms automatically



"AI-Powered Everything"

Every screen has an AI call. No task defined. No numbers. No fallback.



Raw ChatGPT on every button

Breaks without internet. Expensive. Unpredictable. No evaluation possible.

HUMAN-IN-THE-LOOP

When AI confidence is below your threshold → route to human review. Say this explicitly. It's a strength, not a weakness.

A brilliant idea that crashes during the demo is remembered as a failure. Plan for this.

 GitHub Setup Descriptive repo name README with demo link Clean branch structure Commit messages that make sense	 Build Core flow only first Env variables in .env Error states handled Loading indicators present	✓ Test Full demo flow 3× end-to-end On 3 different devices With mobile hotspot (not venue WiFi) Check every edge case in the PS	↑ Deploy Live Vercel · Railway · Render Live URL working and stable UptimeRobot monitoring No last-minute pushes	 Pre-load Data 5–10 realistic fake users Real-looking names, dates, IDs All demo scenarios pre-seeded AI outputs pre-cached	Backup Ready Screen recording saved offline Hotspot from a second phone No new code from Hour 16 Everyone knows the demo script
---	--	---	--	--	--

NON-NEGOTIABLE BEFORE THE FINALE STARTS

- ☐ App deployed on a public live URL that loads in under 3 seconds
- ☐ Database has realistic pre-loaded data — no "Add Test Data" buttons visible to judges
- ☐ Full demo tested on a device that isn't your development machine
- ☐ Screen recording of a flawless demo run saved on a separate offline device
- ☐ GitHub README: project description + live demo link + architecture image + setup steps

The patterns are predictable. Study both.

The same mistakes eliminate teams every year. The same disciplines win every year. Neither is a secret — they're just rarely taken seriously until it's too late.

SIH 2024 WINNERS

SUPERNOVAZ (AYUSH · SIH1556) · Team Coding_Wizard (GAIL · SIH1707) · Hridayam (DST · SIH1547) — different domains, same execution discipline.

TEAMS THAT LOSE

× Vague PS understanding

"We're solving the problem of..." with no named user, no named constraint, no evidence

× 10 features, 3 work

Unstable demo. Judge can't find the core value in the clutter.

× Zero user evidence

"We think users will find it useful" — no interviews, no data, no research

× Empty demo or crash

"Let me add some test data." — fatal, every time

TEAMS THAT WIN

✓ Named user, named constraint

"GAIL field workers face X because Y — the PS itself specifies the 200m geofence requirement."

✓ 3 features, all flawless

One complete journey + one edge case. Demoted in 4 minutes. Nothing breaks.

✓ Evidence-backed claims

"We spoke to 4 people who work in field roles similar to GAIL's. We found..."

✓ Flawless demo + backup

Pre-loaded data. Screen recording ready. Works on the judge's device.

SIH is a launchpad. What you do next is the real project.



Startup Path

Apply to Atal Innovation Mission (AIM), DPIIT Startup India, or your nearest IIM/IIT incubator. Your SIH project has government-level problem validation — that's a genuine advantage in any accelerator application.



Portfolio

Write a proper case study: problem → research → design decisions → technical architecture → outcome. Post it on GitHub and LinkedIn. This is your most credible placement project — external validation from a national-level competition.



Research Paper

SIH gives you a real problem, real data, and a working prototype. IEEE, ACM, and IJCAI publish student research. Many colleges also have internal journals. The evidence is already there — the paper writes itself.



Patent

If your solution introduces a novel method or algorithm, file a provisional patent via your college's Technology Transfer Office (TTO). Takes 2–4 weeks. Costs little. Protects your work before you publish or open-source.



Open Source

Clean up the repo, write good documentation, release publicly. Government organisations and NGOs may adopt it directly. This is how SIH projects sometimes end up in actual production — someone finds them, forks them, deploys them.



Career Capital

"SIH Finalist" or "SIH Winner" on your resume is a real signal. In placement interviews, tell the story: what problem you solved, for which ministry, what you built, and what it does. Keep it under 2 minutes. Practice it.

That's the *full system.*

From understanding the funnel to picking the right PS, building a focused MVP, and presenting with confidence — you now know how the game works.



The next video

Will drop as soon as SIH problem statements are released on the portal.
We'll walk through real PSs — live, together — and help you pick the right one for your team.

49,892

teams started SIH 2024

1,405

reached the finale

You

now know how the other 1,405 did it